20250407EB_MN (celltiter glo, ATP, NCT, GC, RNA)

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2025年4月7日 星期一

Use **2.5mM T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A) for differentiation and experiment.** i didn't add extra Na pyruvate.

Cells didn't grow very well.

split iPSCs into 2 full 6-well plates on the Friday before differentiation

differentiate 5 of 6-wells and all plates with 10mL T2 in cell-repellent plates

split 1 well into 3 wells of iPSCs

2025年4月9日 星期三

T2 -> T3

2025年4月11日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年4月14日 星期一

T4 -> T5

2025年4月16日 星期三

T5 -> T6

2025年4月18日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年4月21日 星期一

T6 -> T7

Dissociation (combined two plates together)

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).

5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

Table4

	A	B
1	cell	passage
2	WT11	P16+

Table1

	A	B
1	run1	volume (mL)
2	trypsin	6
3	HI-FBS/CTWM	12
4	run2	*take some undissolved EB
5	trypsin	2
6	HI-FBS/CTWM	4

Table3

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	4.26	99	7	29.82	A	96	60	36	507
3					B	24	250	16	939
4					C,D	6	1972	12	the rest

plate A

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												
G	coated with PLO+laminin											
H												

plate B

<table>
<tr><th></th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr>
<tr><td>A</td><td rowspan="3">no virus</td><td rowspan="3">vLH92 (2uL/well)</td><td rowspan="3">vLH110 (3uL/well)</td><td rowspan="3">vlH111 (3uL/well)</td><td></td><td></td></tr>
<tr><td>B</td><td></td><td></td></tr>
<tr><td>C</td><td></td><td></td></tr>
<tr><td>D</td><td>new vLH92 virus 0.5uL</td><td>new vLH92 virus 1uL</td><td>new vLH92 virus 3uL</td><td>new vLH92 virus 5uL</td><td></td><td></td></tr>
</table>

plate C,D

	1	2	3
A			
B			

2025年4月22日 星期二

Change T7 -> T7 if added virus

2025年4月23日 星期三

T7-> T7 if not added virus

2025年4月24日 星期四

T7 -> T8 (new recipe from no-glu N2B27 media) for all

***forgot to change media on Saturday

2025年4月28日 星期一

Cells on plate B didn't look very healty (many dead cells attached on the cluster)

The most left column of the plate A died mostly (column 7)

plate C,D looked fine

Change media (T8)

change media

	A	B	C
1	plate	take out	put in
2	A	50	100
3	B	200	400
4	C,D	700	1000

2025年4月29日 星期二

Prepare 25mL 0.15mL from no-glucose T8

prepare media with heavy glucose

	A	B	C	D	E	F	G	H	I
1			stock (mM)	targetconc. (mM)	dilution factor	final volume (mL)	added volume (uL)	media volume (uL)	
2	no-glu T8	C13-glucose	100	2.5	40	5	125	4875	
3			100	0.15	666.6666666667	5	8	4993	

2025年4月30日 星期三

- [x] PlateA: CellTiter Glo assay (at 9:30 am)

*all the treatments are 1h only

prepare ATP calibration

	A	B	C	D	E	F	G
1	A	B	C	D	E	F	G
2	100mM	1mM	1uM	300nM	100nM	30nM	10nM
3	dilution factor from the previous one	100	1000		10		10
4					from C		fromE
5		1uL A in 99uL media	1uL B in 999uL	150uL in 350uL media	50uL C in 450uL media	150uL in 350uL media	50uL D in 450uL media

prepare media with different conc. of glucose

	A	B	C	D	E	F
1	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume (uL)
2	1100	20	55	2000	36	1964
3	20	7.5	2.6666666667	500	188	313
4	20	2.5	8	500	63	438
5	20	0.5	40	500	13	488
6	20	0.15	133.3333333333	500	4	496

plate A1

	1	2	3	4	5	6	7	8	9	10	11	12
A							died	20mM				
B								7.5mM				
C							died	2.5mM				
D							died	0.5mM				
E							died	0.15mM				
F							died	0mM				
G				C			D			no-glu media		
H				E			F			G		

total wells: 48 wells -> 5mL max

Result: there is not much difference between 0mM vs. 2.5mM

Change media: 1/2 T8 -> 1/2 T8

change media1

	A	B	C
1	plate	take out	put in
2	B	200	300
3	D	700	800

Treat plate C at 11:00 am

plate C treatment at 11:00 am

	1	2	3
A	2.5 mM C12-glu 24h	2.5 mM C13-glu 24h	
B	0.15 mM C12-glu 24h	0.15 mM C13-glu 24h	

Treat plate D at 3:45 pm

Prepare MG-132 media

MG-132 dilution

	A	B	C	D	E	F
1	diluted from (uM)	targetconc. (uM)	dilution factor	final volume (uL)	added volume (uL)	media volume (uL)
2	10500	50	210	210	1	209
3	50	1.5	33.3333333333	2000	60	1940

plate D treatment at 3:30 pm

	1	2	3
A	2.5 mM glu 1h	**2.5 mM glu 24h + DMSO**	**2.5 mM glu 24h**
B	0.15 mM glu 1h	**2.5 mM glu 24h + 1.5 uM MG132**	**0.15 mM glu 24h**

2025年5月1日 星期四

GC-MS (plate C)

- [x] Extract GC-MS sample at 10:45 am

plate D treatment at 3:30 pm2

	1	2	3
A	(A) 2.5 mM glu 24h C12	**(B) 2.5 mM glu 24h C13**	**(C) 2.5 mM glu 24h C13**
B	(D) 0.15 mM glu 24h C12	**(E) 0.15 mM glu 24h C13**	**(F) 0.15 mM glu 24h C13**

(unsure)

M1: 0mM

M2: 0.15mM

M3: 2.5mM

RNA extraction (plate D)

- [x] Treat plate D at 2:45 pm
- [x] collect RNA at 3:55 pm

plate D treatment at 3:30 pm1

	1	2	3
A	2.5 mM glu 1h	**2.5 mM glu 24h + DMSO**	**2.5 mM glu 24h**
B	0.15 mM glu 1h	**2.5 mM glu 24h + 1.5 uM MG132**	**0.15 mM glu 24h**

[x] Imaging (plate B)

All the media is no-glu N2B27 media with full supplement (T8 supp and N2, B27)

imaging condition

	1	2	3	4	5	6
A	1	1	1	1		
B	2	2	2	2		
C	3	3	3	3		
D	new vLH92 virus 0.5uL	new vLH92 virus 1uL	new vLH92 virus 3uL	new vLH92 virus 5uL		

Table2

	A	B	C	D	E	F	G
1	condition #	glucose conc. (mM)	volume	FCCP (1000x)	2CG (333x)	volume with SirDNA	SirDNA (5000x)
2	1	2.5	4000			2500	0.5
3	2	0	4000			2500	0.5
4	3	0	4000	4	12	2500	0.5

vLH92 (2ABC)
10 positions/well

O.C. , 40X, zoom1.5, 600x900, 16-bit

	A	B	C	D
1	channel	bit	power (%)	exposure time
2	488	16	30	300
3	Dapi-ex-Green-em	16	30	300

treat three condition one after another. Run all condition at the same time.
take pretreatment images at 5:20 pm (NT): 3 loops, 10min time interval, 20 min total
take cells at 6:16-18 pm, and take postreatment at 6:28 pm (no-glu): 7 loops, 10 min time interval, start 10 min after treatment
note:

- There is 40 min gap between NT and no-glu)
- 10 positions/well, but there are redundant images in NT: position 5 == position 6; position 22 == position 23. I changed position 5 and position 22 into different field. --> remember to remove position 6 and position 23 in NT

vLH110 (3ABC)

vLH111 (4ABC)

O.C. , 60X, 600x900, 12-bit

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	640	12	30	300 ms	110_003 is 30% 100ms
3	561	12	10	100 ms	110_001 is 20%, 200ms
4	488	12	50	1s	

Table5

	A	B	C	D	E	F
1			110 (B)	110 (B)	111 (C)	111 (C)
2	number	condition	treatment time	image 1h	treatment time	image 1h
3	1	2.5mM	7:45	8:45	9:45	10:40
4	2	0mM	8:15	9:23	10:20	11:15
5	3	0mM+FCCP+2DG	8:40	9:55	10:40	11:44

Imaging note:

Tried JOBS module the first time. I am still trying to understand the setting, so some issues happened

vLH110:

- use time loop + point loop + capture as described in the tutorial --> find out using ND acquisition is much easier --> use that for 111
- A3 (condition 1):
  - didn't save the pre-activation files, but I had some test file and I will use them as pre-activation files.
  - combine two 2 time point pre-activation test file into one --> remove the first time point to make a three time frame file.
- I set up the time lapse by time, but it turned out to only have 30 loops for activation file.
- B3 (condition 2):
  - I forgot to change the OC setting of 561 back, so the 561 for it is 30% 100ms.
  - 1 position is out of focuse (although I checked them one by one, but I didn't see which one had problem...

vLH111:

- use ND acquisition
- Still keep activation 30 frames

Is it possible to see live images from JOBS?

2025年5月2日 星期五

Imaging Plate B D123

D1-D4 all express well amount of GFP. However, cells in D1 and D2 (0.5uL and 1 uL) looked healthier than others, especially D4 died a lot.

Perform experiment on D123.

Row D condition

	1	2	3	4	5	6
A	1	1	1	1		
B	2	2	2	2		
C	3	3	3	3		
D	3	2	1			

vLH92 (D1-3)

10 positions/well

O.C. , 40X, zoom1.5, 600x900, 16-bit1

	A	B	C	D
1	channel	bit	power (%)	exposure time
2	488	16	25	300
3	Dapi-ex-Green-em	16	25	300

treat three condition one after another. Run all condition at the same time.

procedure (setup a JOBS module to run NT first and wait for drug treatment, and then do another acquisition 10 min later)

1. take pretreatment images at 5:20 pm (NT): 3 loops, 10min time interval, 20 min total
2. remove media
3. add ~ 500uL media for wash
4. add 600/700uL media
5. click OK after adding all of them.
6. the program will wait for 10min
7. take post-treatment acquisition(no-glu): 7 loops, 10 min time interval, start 10 min after treatment.

Note: the position 3 after treatment was drifted out of the field. -> do not use.

2025年5月16日 星期五

Practice how to analyze GC data

From the Openchrom, we will get three tables: **(I) Retention time, (II) Total ion counts, (III) Mass isotopomer abundances**

1. Quality check: RT check using (I) **Retention Times (min.).** Max - Min. The difference should not be more than 0.2 min

- [x] Calculate **(A) full molecule isotopomer abundances** with existed fraction
    a. Remove value <0 from (III) **Mass isotopomer abundances** table.

b. Existed fraction: Sum(true labeled signal of the moleculue)/Sum(all fraction)
c. Do not need to include theory column (which is the natural abundance)
d. Use the following table to pick the true signal.

Table7

	A	B	C	D
1	**metabolites**	**Max M**	**True label M**	**internal standard**
2	**Pyr_ 174**	8	3	Ala_264 (M4)
3	**Lac_ 261**	8	3	Ala_264 (M4)
4	**Ala_ 260**	8	3	Ala_264 (M4)
5	**Gly_ 246**	6	2	Gly_249 (M3)
6	**Val_ 288**	8	5	Val_294 (M6)
7	**Leu_ 274**	8	5	Leu_280 (M6)
8	**Ile_ 274**	8	M6 (should be M5?)	Ile_280 (M6)
9	**Suc_ 289**	7	4	Ile_280 (M6)
10	**Fum_ 287**	5	4	Ile_280 (M6)
11	**Ser_ 390**	8	3	Ser_394 (M4)
12	**Thr_ 404**	6	4	Thr_409 (M5)
13	**Met_ 320**	8	5	Met_326 (M6)
14	**Akg_ 346**	9	5	Met_326 (M6)
15	**Mal_ 419**	9	4	Phe_346 (M10)
16	**Phe_ 336**	10	9	Phe_346 (M10)
17	**Asp_ 418**	10	4	Asp_423 (M5)
18	**Pro_ 330**	10	5	Pro_336 (M6)
19	**Glu_ 432**	10	5	Glu_438 (M6)
20	**Lys_ 431**	12	6	Lys_439 (M8)
21	**Gln_ 431**	10	5	Tyr_305 (M3)
22	**Cit_ 591**	9	6	Tyr_305 (M3)
23	**Tyr_ 302**	8	0-2;4-6	Tyr_305 (M3)
24	**His_ 440**	12	6	His_449 (M9)

- [x] **(B) Fractional labeling of selected isotopmers**

a. individual metabolites abundance/Sum(true labeled signal of the moleculue) from (A)
b. If calculation returns ERROR, return 0

- [x] **(C) Total ion counts of selected existed isotopmers**

  - Existed fraction from (A) * Total ion counts (II)
- [x] **(D) Relative abundance of internal standard**
  - individual metabolites abundance from (A) * total ion count (II) of the internal standard (Table 7)
- [x] **(E) Relative abundance of selected isotopmers**
  - **(B)** * **(C)**
- [x] **(F) Absolute abundance of selected isotopmers/pmol**
  - **(E)** / **(D)** * amount of internal standard (mol)
  - internal standard concentration is 0.5uM (Cambridge Isotope Lab, MSK-A2) in MeOH
  - If it's chloroform extraction:
    - 180 (volume we take after centrifuge) * 0.5 * 200 (MeOH mix volume we added)/(200 (MeOH) +100 (H2O)
    - 180*0.5*200/300 = 60
  - If it's media extraction:
    - 180 (volume we take after centrifuge) * 0.5 * 190 (MeOH mix) * (190+10 (H2O))
    - 180*0.5*190/200 = 85.5

Plot

- [ ] Relative internal standard in every sample
- [ ]

questions for Zixuan:

1. from Data(040425): RT check is using =MAX(A3:B3)-MIN(B3:AO3) instead of =MAX(B3:AO3)-MIN(B3:AO3). is there a reason?
2. Ile can be labeled with 6 C, but the internal control is also M6? --> true: M0-M5
3. Tryrosin exclude M3 -> yes
4.